

Draft

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1 Source term analysis

1.1 Electric field source terms

Use the conventions $\delta_x^{E,2} = \delta_x^H \delta_x^E$ and $\delta_x^{H,2} = \delta_x^E \delta_x^H$. Let $S_{E,1}^n$ be the electric field source term in between n and $n + \frac{1}{2}$, and $S_{E,2}^n$ be that in between $n + \frac{1}{2}$ and $n + 1$.

$$(1 - \gamma \delta_x^{E,2}) E_z^{n+\frac{1}{2}} = E_z^n + C_b \delta_x^H H_y^n + S_{E,1}^n \quad (1)$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b \delta_x^E E_z^{n+\frac{1}{2}} \quad (2)$$

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^{n+\frac{1}{2}} + S_{E,2}^n \quad (3)$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b \delta_x^E E_z^{n+\frac{1}{2}} \quad (4)$$

Shifting back equations (3) and (4) one full step gives

$$E_z^n = E_z^{n-\frac{1}{2}} + C_b \delta_x^H H_y^{n-\frac{1}{2}} + S_{E,2}^{n-1} \quad (5)$$

$$H_y^n = H_y^{n-\frac{1}{2}} + D_b \delta_x^E E_z^{n-\frac{1}{2}} \quad (6)$$

To eliminate H_y from equations (1)–(3), substitute equation (2) into equation (3):

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^n + \gamma \delta_x^{E,2} E_z^{n+\frac{1}{2}} + S_{E,2}^n. \quad (7)$$

Equation (1) gives

$$C_b \delta_x^H H_y^n = (1 - \gamma \delta_x^{E,2}) E_z^{n+\frac{1}{2}} - E_z^n - S_{E,1}^n. \quad (8)$$

Using $\gamma \delta_x^{E,2} = C_b D_b \delta_x^H \delta_x^E$, equations (7) and (8) reduce to

$$E_z^{n+\frac{1}{2}} = \frac{1}{2} (E_z^{n+1} + E_z^n + S_{E,1}^n - S_{E,2}^n). \quad (9)$$

Shifting back one full step gives

$$E_z^{n-\frac{1}{2}} = \frac{1}{2} (E_z^n + E_z^{n-1} + S_{E,1}^{n-1} - S_{E,2}^{n-1}). \quad (10)$$

Next, eliminate H_y using equations (1), (5), and (6). Applying $C_b \delta_x^H$ to both sides of equation (6) gives

$$C_b \delta_x^H H_y^n = C_b \delta_x^H H_y^{n-\frac{1}{2}} + \gamma \delta_x^{E,2} E_z^{n-\frac{1}{2}}. \quad (11)$$

Eliminate $C_b \delta_x^H H_y^{n-\frac{1}{2}}$ by equation (5) gives

$$C_b \delta_x^H H_y^n = E_z^n - (1 - \gamma \delta_x^{E,2}) E_z^{n-\frac{1}{2}} - S_{E,2}^{n-1}. \quad (12)$$

Substituting into equation (1) to eliminate H_y :

$$(1 - \gamma \delta_x^{E,2}) (E_z^{n+\frac{1}{2}} + E_z^{n-\frac{1}{2}}) = 2E_z^n + S_{E,1}^n - S_{E,2}^{n-1}. \quad (13)$$

Finally, substitute equations (9) and (10) into equation (13) to eliminate the half-step values of E_z :

$$(1 - \gamma \delta_x^{E,2}) E_z^{n+1} = 2 (1 + \gamma \delta_x^{E,2}) E_z^n - (1 - \gamma \delta_x^{E,2}) E_z^{n-1} + (1 + \gamma \delta_x^{E,2}) S_{E,1}^n - (1 - \gamma \delta_x^{E,2}) S_{E,1}^{n-1} + (1 - \gamma \delta_x^{E,2}) S_{E,2}^n - (1 + \gamma \delta_x^{E,2}) S_{E,2}^{n-1}. \quad (14)$$

Recall that $\gamma = C_b D_b = c^2 \Delta t^2 / 4$, then equation (14) results in,

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{1}{\Delta t^2} \left[(1 + \gamma \delta_x^{E,2}) S_{E,1}^n - (1 - \gamma \delta_x^{E,2}) S_{E,1}^{n-1} + (1 - \gamma \delta_x^{E,2}) S_{E,2}^n - (1 + \gamma \delta_x^{E,2}) S_{E,2}^{n-1} \right]. \quad (15)$$

Suppose the same source is applied in both substeps, with $S_{E,1}^n = S_{E,2}^n = \frac{\Delta t}{2} S^{n+\frac{1}{2}}$ and $S_{E,1}^{n-1} = S_{E,2}^{n-1} = \frac{\Delta t}{2} S^{n-\frac{1}{2}}$. Then the source terms in equation (15) reduce to

$$\frac{\Delta t}{2} \left[(1 + \gamma \delta_x^{E,2}) S^{n+\frac{1}{2}} + (1 - \gamma \delta_x^{E,2}) S^{n+\frac{1}{2}} - (1 - \gamma \delta_x^{E,2}) S^{n-\frac{1}{2}} - (1 + \gamma \delta_x^{E,2}) S^{n-\frac{1}{2}} \right] = \Delta t \left(S^{n+\frac{1}{2}} - S^{n-\frac{1}{2}} \right). \quad (16)$$

Therefore equation (15) becomes

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{S^{n+\frac{1}{2}} - S^{n-\frac{1}{2}}}{\Delta t}. \quad (17)$$

Assume $S^{n+a} = S(t_n + a\Delta t)$ and $F^n = S_t(t_n)$. The source approximation in equation (17) has the Taylor expansion

$$\frac{S^{n+\frac{1}{2}} - S^{n-\frac{1}{2}}}{\Delta t} = F^n + \frac{\Delta t^2}{24} S_{ttt}^n + O(\Delta t^4), \quad \text{error} = \frac{\Delta t^2}{24} S_{ttt}^n + O(\Delta t^4). \quad (18)$$

Thus the midpoint-source case is second-order accurate for $F^n = S_t(t_n)$.

Alternatively, suppose the first source is evaluated at the first quarter-step and the second source at the third quarter-step: $S_{E,1}^n = \frac{\Delta t}{2} S^{n+\frac{1}{4}}$, $S_{E,2}^n = \frac{\Delta t}{2} S^{n+\frac{3}{4}}$, $S_{E,1}^{n-1} = \frac{\Delta t}{2} S^{n-\frac{3}{4}}$, and $S_{E,2}^{n-1} = \frac{\Delta t}{2} S^{n-\frac{1}{4}}$. Then equation (15) becomes

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{1}{2\Delta t} \left[(1 + \gamma \delta_x^{E,2}) S^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{E,2}) S^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{E,2}) S^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{E,2}) S^{n-\frac{1}{4}} \right]. \quad (19)$$

The source approximation in equation (19) has the Taylor expansion

$$F^n - \frac{\gamma}{2} \delta_x^{E,2} F^n + \frac{7\Delta t^2}{96} S_{ttt}^n + O(\Delta t^4), \quad \text{error} = -\frac{\gamma}{2} \delta_x^{E,2} F^n + \frac{7\Delta t^2}{96} S_{ttt}^n + O(\Delta t^4). \quad (20)$$

Since $\gamma = O(\Delta t^2)$, the quarter-step source case is also second-order accurate.

Assume $S(x, t) = \hat{S} e^{i(kx + \omega t)}$, and define $\theta = \omega \Delta t$, $\delta_x^{E,2} e^{ikx} = \lambda_x e^{ikx}$, and $q = -\gamma \lambda_x \geq 0$. Substituting into the source term on the right-hand side of equation (17) gives

$$\begin{aligned} F_{\text{mid}}^n &= \frac{S^{n+\frac{1}{2}} - S^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{\hat{S} e^{i(kx + \omega t_n)}}{\Delta t} \left(e^{i\theta/2} - e^{-i\theta/2} \right) \\ &= \frac{2i\hat{S}}{\Delta t} \sin\left(\frac{\theta}{2}\right) e^{i(kx + \omega t_n)}, \end{aligned} \quad (21)$$

so $\hat{F}_{\text{mid}} = \frac{2i\hat{S}}{\Delta t} \sin\left(\frac{\theta}{2}\right)$. Substituting into the source term on the right-hand side of equation (19) gives

$$\begin{aligned} F_{\text{quarter}}^n &= \frac{1}{2\Delta t} \left[(1 + \gamma \delta_x^{E,2}) S^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{E,2}) S^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{E,2}) S^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{E,2}) S^{n-\frac{1}{4}} \right] \\ &= \frac{\hat{S} e^{i(kx + \omega t_n)}}{2\Delta t} \left[(1 - q) e^{i\theta/4} - (1 + q) e^{-3i\theta/4} + (1 + q) e^{3i\theta/4} - (1 - q) e^{-i\theta/4} \right] \\ &= \frac{i\hat{S}}{\Delta t} \left[(1 - q) \sin\left(\frac{\theta}{4}\right) + (1 + q) \sin\left(\frac{3\theta}{4}\right) \right] e^{i(kx + \omega t_n)}. \end{aligned} \quad (22)$$

so $\hat{F}_{\text{quarter}} = \frac{i\hat{S}}{\Delta t} \left[(1 - q) \sin\left(\frac{\theta}{4}\right) + (1 + q) \sin\left(\frac{3\theta}{4}\right) \right]$. Therefore the magnitude ratio between the quarter-step and midpoint-source terms is

$$\frac{|\hat{F}_{\text{quarter}}|}{|\hat{F}_{\text{mid}}|} = \left| \cos\left(\frac{\theta}{4}\right) + \frac{q \cos\left(\frac{\theta}{2}\right)}{2 \cos\left(\frac{\theta}{4}\right)} \right|. \quad (23)$$

For a spatially uniform source, $q = 0$, so the quarter-step magnitude is smaller by the factor

$$\frac{|\hat{F}_{\text{quarter}}|}{|\hat{F}_{\text{mid}}|} = \left| \cos\left(\frac{\omega \Delta t}{4}\right) \right|. \quad (24)$$

1.2 Magnetic field source terms

Let $S_{H,1}^n$ be the magnetic field source term in between n and $n + \frac{1}{2}$, and $S_{H,2}^n$ be that in between $n + \frac{1}{2}$ and $n + 1$.

$$(1 - \gamma\delta_x^{E,2}) E_z^{n+\frac{1}{2}} = E_z^n + C_b\delta_x^H H_y^n. \quad (25)$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b\delta_x^E E_z^{n+\frac{1}{2}} + S_{H,1}^n. \quad (26)$$

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b\delta_x^H H_y^{n+\frac{1}{2}}. \quad (27)$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b\delta_x^E E_z^{n+\frac{1}{2}} + S_{H,2}^n. \quad (28)$$

Shifting back equations (27) and (28) one full step gives

$$E_z^n = E_z^{n-\frac{1}{2}} + C_b\delta_x^H H_y^{n-\frac{1}{2}}. \quad (29)$$

$$H_y^n = H_y^{n-\frac{1}{2}} + D_b\delta_x^E E_z^{n-\frac{1}{2}} + S_{H,2}^{n-1}. \quad (30)$$

To eliminate E_z from equations (26)–(28), equation (26) gives

$$D_b\delta_x^E E_z^{n+\frac{1}{2}} = H_y^{n+\frac{1}{2}} - H_y^n - S_{H,1}^n. \quad (31)$$

Substituting into equation (28) gives

$$H_y^{n+1} = 2H_y^{n+\frac{1}{2}} - H_y^n - S_{H,1}^n + S_{H,2}^n. \quad (32)$$

Solving for the half-step value gives

$$H_y^{n+\frac{1}{2}} = \frac{1}{2} (H_y^{n+1} + H_y^n + S_{H,1}^n - S_{H,2}^n). \quad (33)$$

Shifting back one full step gives

$$H_y^{n-\frac{1}{2}} = \frac{1}{2} (H_y^n + H_y^{n-1} + S_{H,1}^{n-1} - S_{H,2}^{n-1}). \quad (34)$$

Next, eliminate E_z using equations (25), (26), (29), and (30). Applying $D_b\delta_x^E$ to both sides of equation (29) gives

$$D_b\delta_x^E E_z^n = D_b\delta_x^E E_z^{n-\frac{1}{2}} + \gamma\delta_x^{H,2} H_y^{n-\frac{1}{2}}. \quad (35)$$

Eliminate $D_b\delta_x^E E_z^{n-\frac{1}{2}}$ by equation (30) gives

$$D_b\delta_x^E E_z^n = H_y^n - (1 - \gamma\delta_x^{H,2}) H_y^{n-\frac{1}{2}} - S_{H,2}^{n-1}. \quad (36)$$

Applying $D_b\delta_x^E$ to both sides of equation (25) gives

$$(1 - \gamma\delta_x^{H,2}) D_b\delta_x^E E_z^{n+\frac{1}{2}} = D_b\delta_x^E E_z^n + \gamma\delta_x^{H,2} H_y^n. \quad (37)$$

Eliminate $D_b\delta_x^E E_z^{n+\frac{1}{2}}$ by equation (26) gives

$$D_b\delta_x^E E_z^n = (1 - \gamma\delta_x^{H,2}) H_y^{n+\frac{1}{2}} - H_y^n - (1 - \gamma\delta_x^{H,2}) S_{H,1}^n. \quad (38)$$

Equating equations (38) and (36) eliminates E_z^n :

$$(1 - \gamma\delta_x^{H,2}) (H_y^{n+\frac{1}{2}} + H_y^{n-\frac{1}{2}}) = 2H_y^n + (1 - \gamma\delta_x^{H,2}) S_{H,1}^n - S_{H,2}^{n-1}. \quad (39)$$

Finally, substitute equations (33) and (34) into equation (39) to eliminate the half-step values of H_y :

$$(1 - \gamma\delta_x^{H,2}) H_y^{n+1} = 2(1 + \gamma\delta_x^{H,2}) H_y^n - (1 - \gamma\delta_x^{H,2}) H_y^{n-1} + (1 - \gamma\delta_x^{H,2}) S_{H,1}^n - (1 - \gamma\delta_x^{H,2}) S_{H,1}^{n-1} + (1 - \gamma\delta_x^{H,2}) S_{H,2}^n - (1 + \gamma\delta_x^{H,2}) S_{H,2}^{n-1}. \quad (40)$$

Recall that $\gamma = C_b D_b = c^2 \Delta t^2 / 4$, then equation (40) results in,

$$\begin{aligned} \frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = \\ c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4} \right) + \frac{1}{\Delta t^2} \left[(1 - \gamma\delta_x^{H,2}) S_{H,1}^n - (1 - \gamma\delta_x^{H,2}) S_{H,1}^{n-1} + (1 - \gamma\delta_x^{H,2}) S_{H,2}^n - (1 + \gamma\delta_x^{H,2}) S_{H,2}^{n-1} \right]. \end{aligned} \quad (41)$$

Suppose the same source is applied in both substeps, with $S_{H,1}^n = S_{H,2}^n = \frac{\Delta t}{2} S_H^{n+\frac{1}{2}}$ and $S_{H,1}^{n-1} = S_{H,2}^{n-1} = \frac{\Delta t}{2} S_H^{n-\frac{1}{2}}$. Then the source terms in equation (41) reduce to

$$\frac{\Delta t}{2} \left[(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{2}} + (1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{2}} - (1 - \gamma \delta_x^{H,2}) S_H^{n-\frac{1}{2}} - (1 + \gamma \delta_x^{H,2}) S_H^{n-\frac{1}{2}} \right] = \Delta t \left[(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}} \right]. \quad (42)$$

Therefore equation (41) becomes

$$\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4} \right) + \frac{(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t}. \quad (43)$$

Assume $S_H^{n+a} = S_H(t_n + a\Delta t)$ and $F_H^n = (S_H)_t(t_n)$. The source approximation in equation (43) has the Taylor expansion

$$F_H^n - \frac{\gamma}{\Delta t} \delta_x^{H,2} S_H^n - \frac{\gamma}{2} \delta_x^{H,2} F_H^n + \frac{\Delta t^2}{24} (S_H)_{ttt}^n + O(\Delta t^3). \quad (44)$$

Since $\gamma = O(\Delta t^2)$, the term $-\gamma \delta_x^{H,2} S_H^n / \Delta t$ is $O(\Delta t)$. Thus this midpoint-source case is first-order accurate for a spatially varying S_H , but remains second-order when $\delta_x^{H,2} S_H = 0$.

Alternatively, suppose the first source is evaluated at the first quarter-step and the second source at the third quarter-step: $S_{H,1}^n = \frac{\Delta t}{2} S_H^{n+\frac{1}{4}}$, $S_{H,2}^n = \frac{\Delta t}{2} S_H^{n+\frac{3}{4}}$, $S_{H,1}^{n-1} = \frac{\Delta t}{2} S_H^{n-\frac{3}{4}}$, and $S_{H,2}^{n-1} = \frac{\Delta t}{2} S_H^{n-\frac{1}{4}}$. Then equation (41) becomes

$$\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4} \right) + \frac{1}{2\Delta t} \left[(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{H,2}) S_H^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{H,2}) S_H^{n-\frac{1}{4}} \right]. \quad (45)$$

The source approximation in equation (45) has the Taylor expansion

$$F_H^n - \frac{\gamma}{\Delta t} \delta_x^{H,2} S_H^n - \frac{3\gamma}{4} \delta_x^{H,2} F_H^n + \frac{7\Delta t^2}{96} (S_H)_{ttt}^n + O(\Delta t^3). \quad (46)$$

Thus the quarter-step source case is also first-order accurate for a spatially varying S_H , and second-order when $\delta_x^{H,2} S_H = 0$.

Assume $S_H(x, t) = \hat{S}_H e^{i(kx + \omega t)}$, and define $\theta = \omega \Delta t$, $\delta_x^{H,2} e^{ikx} = \lambda_{H,x} e^{ikx}$, and $q_H = -\gamma \lambda_{H,x} \geq 0$. Substituting into the source term on the right-hand side of equation (43) gives

$$\begin{aligned} F_{H,\text{mid}}^n &= \frac{(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{\hat{S}_H e^{i(kx + \omega t_n)}}{\Delta t} \left[(1 + q_H) e^{i\theta/2} - e^{-i\theta/2} \right] \\ &= \frac{\hat{S}_H}{\Delta t} \left[q_H \cos\left(\frac{\theta}{2}\right) + i(2 + q_H) \sin\left(\frac{\theta}{2}\right) \right] e^{i(kx + \omega t_n)}. \end{aligned} \quad (47)$$

so $\hat{F}_{H,\text{mid}} = \frac{\hat{S}_H}{\Delta t} [q_H \cos(\frac{\theta}{2}) + i(2 + q_H) \sin(\frac{\theta}{2})]$. Substituting into the source term on the right-hand side of equation (45) gives

$$\begin{aligned} F_{H,\text{quarter}}^n &= \frac{1}{2\Delta t} \left[(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{H,2}) S_H^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{H,2}) S_H^{n-\frac{1}{4}} \right] \\ &= \frac{\hat{S}_H e^{i(kx + \omega t_n)}}{2\Delta t} \left[(1 + q_H) e^{i\theta/4} - (1 + q_H) e^{-3i\theta/4} + (1 + q_H) e^{3i\theta/4} - (1 - q_H) e^{-i\theta/4} \right] \\ &= \frac{\hat{S}_H}{\Delta t} \left[q_H \cos\left(\frac{\theta}{4}\right) + i \left(\sin\left(\frac{\theta}{4}\right) + (1 + q_H) \sin\left(\frac{3\theta}{4}\right) \right) \right] e^{i(kx + \omega t_n)}. \end{aligned} \quad (48)$$

Therefore the magnitude ratio between the quarter-step and midpoint magnetic-source terms is

$$\frac{|\hat{F}_{H,\text{quarter}}|}{|\hat{F}_{H,\text{mid}}|} = \frac{\sqrt{q_H^2 \cos^2(\frac{\theta}{4}) + [\sin(\frac{\theta}{4}) + (1 + q_H) \sin(\frac{3\theta}{4})]^2}}{\sqrt{q_H^2 \cos^2(\frac{\theta}{2}) + (2 + q_H)^2 \sin^2(\frac{\theta}{2})}}. \quad (49)$$

For a spatially uniform source, $q_H = 0$, this reduces to

$$\frac{|\hat{F}_{H,\text{quarter}}|}{|\hat{F}_{H,\text{mid}}|} = \left| \cos\left(\frac{\omega \Delta t}{4}\right) \right|. \quad (50)$$